

Experiment 'p_6_2_c_replace_even_with_zero_array' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_c_replace_even_with_zero_array

Natural language statement: Write a method to replace all even elements in an array with 0.

Method signature: p_6_2_c_replace_even_with_zero_array(inputs: array<int>) returns (result: array<int>)

Ensures

- `result.Length == inputs.Length`
- `forall i :: 0 <= i < inputs.Length ==> (inputs[i] % 2 == 0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to replace all even elements in an array with 0.

The signature should be:

`p_6_2_c_replace_even_with_zero_array(inputs: array<int>) returns (result: array<int>)`

The method should respect the following contract:

`ensures result.Length == inputs.Length, ensures forall i :: 0 <= i < inputs.Length ==> (`
 `↪ inputs[i] % 2 == 0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] ==`
 `↪ inputs[i])`

Produce and show only the Dafny body of this method, including the curly braces that
 `↪` surround it. Do not show the signature nor contract. You must not use 'assume' nor
 `↪` 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0EibEO82irl0si0f3Y4rAXML6Y1

Model: gpt-5.5-2026-04-23

Created at: 1780571352

Finish reason: stop

Usage: 368 tokens in, and 516 tokens out

```
{
  result := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant forall j :: 0 <= j < i ==> (inputs[j] % 2 == 0 ==> result[j] == 0) && (
      ↪ inputs[j] % 2 != 0 ==> result[j] == inputs[j])
  {
    if inputs[i] % 2 == 0 {
      result[i] := 0;
    } else {
      result[i] := inputs[i];
    }
    i := i + 1;
  }
}
```

Final Program

```
method p_6_2_c_replace_even_with_zero_array(inputs: array<int>) returns (result: array<
  ↪ int>)
  ensures result.Length == inputs.Length
  ensures forall i :: 0 <= i < inputs.Length ==> (inputs[i] % 2 == 0 ==> result[i]
    ↪ == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])
{
  result := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant forall j :: 0 <= j < i ==> (inputs[j] % 2 == 0 ==> result[j] == 0) && (
      ↪ inputs[j] % 2 != 0 ==> result[j] == inputs[j])
  {
    if inputs[i] % 2 == 0 {
      result[i] := 0;
    } else {
      result[i] := inputs[i];
    }
    i := i + 1;
  }
}
```

Total Token Usage

Input tokens: 368

Output tokens: 516

Reasoning tokens: 345

Sum of 'total tokens': 884

Experiment Timings

Overall Experiment started at 1780571351985, ended at 1780571382472, lasting 30487ms (30.49 seconds)

Iteration #1 started at 1780571351986, ended at 1780571382472, lasting 30486ms (30.49 seconds)